

Osterquill Toxicology Inc

AP02 reserve analogues: integrated hERG and CYP screen, 44 compounds

Final Report

Study OT-23-S0046 (LIMS OQ/2023/000755)

Client: Avernoelwick Pharmaceuticals Ltd

Study Director: Dr N. Takamori

Report date: March 23, 2023

1 Summary

44 compounds from the Avernoelwick aminopyrimidine regioisomer series 02 were tested for Avernoelwick Pharmaceuticals Ltd in the hERG (IKr) safety pharmacology, CYP safety interaction panel. Testing started on March 8, 2023 and results were reported on March 23, 2023. All runs met control validity criteria. No compound inhibited hERG with an IC50 below 1 μ M. Most compounds were weak or inactive (IC50 at or above 10 μ M, or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected.

Table 1 Study summary

Metric	Value
Project	OT-23-S0046
Client	Avernoelwick Pharmaceuticals Ltd
Compounds tested	44
Series / chemotype	Avernoelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	15
Price	USD 25,000
hERG IC50 <1 μ M	0
hERG IC50 1-10 μ M	19
hERG IC50 $\geq$ 10 μ M	25
hERG median IC50 (μ M)	12
E-4031 control IC50 (nM)	15
CYP1A2 >50% inhibition @10 μ M	2
CYP2C9 >50% inhibition @10 μ M	3
CYP2C19 >50% inhibition @10 μ M	11
CYP2D6 >50% inhibition @10 μ M	8
CYP3A4 >50% inhibition @10 μ M	13
Compounds with any CYP >50%	19
Ketoconazole (3A4) IC50 (μ M)	0.023

Compliance statement

This exploratory in vitro investigation is not conducted under GLP; procedures follow Osterquill standard operating procedures and the investigation is audited by Quality Assurance for data integrity only.

2 Objective

To rank the test article cohort for hERG (IKr) safety pharmacology, CYP safety interaction panel liability so that Avelon Pharmaceuticals Ltd could choose analogues for progression.

3 Test items

44 compounds were received as 10 mM DMSO solutions in one test article cohort on March 8, 2023 and logged against OQ/2023/000755. Identity and concentration were taken as supplied by the client.

4 Methods

hERG (IKr) safety pharmacology

HEK293 cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells were held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final), with 3-5 minutes per concentration.

- Control validity: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current amplitude ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- Positive control E-4031 on every run; IC50 must fall within 5-50 nM
- IC50 from a four-parameter logistic fit of mean tail-current inhibition; $n \geq 2$ cells per compound

CYP safety interaction panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation was measured by LC-MS/MS against vehicle (0.1% DMSO) incubations.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Results are reported as mean percent inhibition relative to vehicle. Values above 50% at 10 μ M are flagged for follow-up; small negative values indicate no inhibition.

5 Results

hERG results

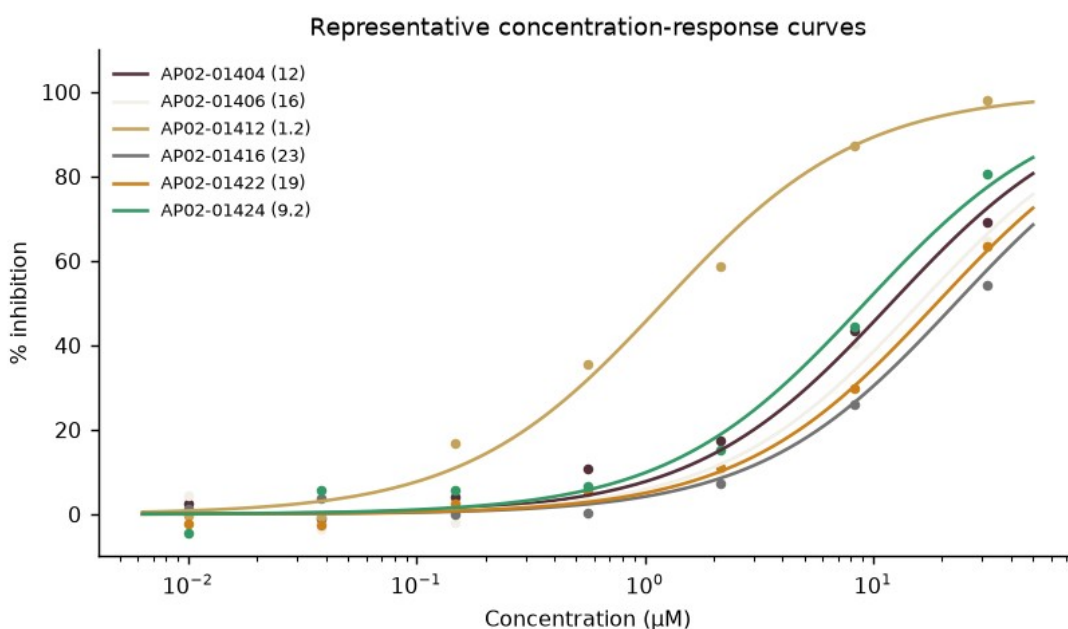
All 44 compounds were tested. No compound inhibited hERG with an IC₅₀ below 1 µM. Most compounds were weak or inactive (IC₅₀ at or above 10 µM, or above the top concentration). The median IC₅₀ was 12 µM (0 below 1 µM, 19 between 1 and 10 µM, 25 at or above 10 µM). E-4031 gave an IC₅₀ of 15 nM, within the acceptance range.

Table 2 hERG IC₅₀ by compound (µM)

Compound	IC ₅₀ rep 1 (µM)	IC ₅₀ rep 2 (µM)	IC ₅₀ rep 3 (µM)	IC ₅₀ mean (µM)	% inhibition @10 µM	n	QC note
AP02-01400	>30	>30		>30	26	2	
AP02-01404	12	6.8	8.5	8.9	52	3	
AP02-01406	16	9.9		13	45	2	
AP02-01412	1.2	1.6		1.4	85	2	
AP02-01416	23	24		24	25	2	
AP02-01422	19	17		18	38	2	
AP02-01424	9.2	11	9.7	10	43	3	
AP02-01428	28	>30		>30	17	2	
AP02-01436	22	16		19	30	2	
AP02-01440	14	11		12	47	2	
AP02-01446	15	14		15	42	2	
AP02-01448	4.1	3.5		3.8	75	2	
AP02-01452	16	22		18	39	2	
AP02-01458	5.0	4.9		4.9	70	2	
AP02-01464	15	14		15	44	2	
AP02-01470	6.1	4.8		5.4	67	2	
AP02-01472	21	13		16	37	2	
AP02-01476	5.7	9.6		7.4	56	2	
AP02-01478	2.5	3.5		3.0	80	2	
AP02-01484	7.7	15		11	47	2	
AP02-01490	>30	26	>30	>30	22	3	
AP02-01494	3.6	2.7		3.1	77	2	
AP02-01496	30	24		27	30	2	
AP02-01500	6.4	4.9		5.6	64	2	
AP02-01502	8.7	9.4		9.1	52	2	
AP02-01508	16	12		14	44	2	
AP02-01514	12	21		16	34	2	
AP02-01518	2.3	2.4		2.4	82	2	
AP02-01520	24	>30		>30	26	2	
AP02-01530	16	14		15	37	2	
AP02-01532	5.4	4.5		5.0	68	2	
AP02-01536	4.4	5.0		4.7	63	2	
AP02-01538	21	11		15	44	2	
AP02-01542	10	6.6	11	9.2	49	3	
AP02-01550	8.2	10		9.2	53	2	
AP02-01554	25	28	>30	27	29	3	
AP02-01556	6.8	7.2		7.0	63	2	

AP02-01560	8.1	6.2		7.1	60	2	
AP02-01566	12	15		13	45	2	Run 2 rejected (E-4031 outside range); repeated
AP02-01568	12	15		13	42	2	
AP02-01572	6.2	4.6		5.3	66	2	
AP02-01574	22	16		19	38	2	
AP02-01584	9.0	11		10	41	2	
AP02-01590	10	8.1		9.0	56	2	Precipitation observed at top concentration

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 15 nM (acceptance 5-50 nM)



CYP inhibition results

Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. In total 19 compounds inhibited at least one isoform by more than 50% at 10 μ M (CYP3A4: 13; CYP2D6: 8; CYP2C19: 11; CYP2C9: 3; CYP1A2: 2). Ketoconazole inhibited CYP3A4 with an IC₅₀ of 0.023 μ M.

Table 3 Percent inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
AP02-01400	13	10	51	22	58
AP02-01404	1	5	9	5	16
AP02-01406	1	-3	9	-1	-3
AP02-01412	38	86	94	45	98
AP02-01416	1	-2	-6	1	7
AP02-01422	-2	0	9	3	-0
AP02-01424	4	-4	4	4	-2
AP02-01428	54	5	43	8	36
AP02-01436	8	1	7	12	-2

AP02-01440	-8	-3	3	14	-1
AP02-01446	15	84	90	6	93
AP02-01448	-6	-1	1	6	68
AP02-01452	3	22	73	10	12
AP02-01458	14	43	77	47	23
AP02-01464	5	1	-3	6	8
AP02-01470	2	5	1	5	-2
AP02-01472	-6	2	5	19	41
AP02-01476	5	19	50	66	39
AP02-01478	27	17	65	76	51
AP02-01484	-1	2	29	1	8
AP02-01490	-3	8	-1	4	2
AP02-01494	46	22	68	14	61
AP02-01496	-11	-0	-1	4	8
AP02-01500	-3	1	1	3	8
AP02-01502	3	-7	16	1	6
AP02-01508	2	43	13	95	72
AP02-01514	-2	9	48	4	7
AP02-01518	6	18	87	89	65
AP02-01520	7	12	20	9	45
AP02-01530	3	0	1	6	16
AP02-01532	2	12	9	68	33
AP02-01536	4	5	80	77	72
AP02-01538	0	1	11	51	-4
AP02-01542	10	8	4	2	-1
AP02-01550	2	-7	17	4	8
AP02-01554	71	48	81	86	89
AP02-01556	-0	69	91	47	73
AP02-01560	-2	3	-2	-5	6
AP02-01566	3	-6	29	3	76
AP02-01568	-3	8	17	35	34
AP02-01572	1	12	18	20	14
AP02-01574	28	0	35	4	87
AP02-01584	-1	2	-6	8	24
AP02-01590	19	13	40	5	47

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.7; sulfaphenazole (CYP2C9) 0.33; ticlopidine (CYP2C19) 1.2; quinidine (CYP2D6) 0.042; ketoconazole (CYP3A4) 0.023

6 Discussion

No compound inhibited hERG with an IC₅₀ below 1 μ M. Most compounds were weak or inactive (IC₅₀ at or above 10 μ M, or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. Results are exploratory and are intended for compound selection; they are not a substitute for a definitive GLP safety pharmacology study.

7 Conclusion

The data support ranking within the set. Recommended next step: select analogues without a hERG or CYP signal for further investigation and repeat any single-run outliers.

Name	Role	Signature	Date
Naomi Takamori	Study Director	<i>Takamori</i>	March 23, 2023
Lucia Moretti	GLP Quality Assurance and Archive Lead	<i>Moretti</i>	March 22, 2023